# Every single beta-reduction step is a parallel beta-reduction step

The Open Logic Project contributors

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-lambda-calculus-church-rosser/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: the shape analysis, the double reflexivity and the closing rule application appear verbatim; the author's juxtaposed rule citations defn:bredpar defn:bredpar4 are delivered as is.

*Editorial note.* The label lem:one-par names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own lem environment in the unit source.

*Editorial note.* OLP reuses this label inside the unit; the card is identified by its statement span (source lines 486-488) and the claim key carries the statement start line.

*Allowed standard prerequisites (editorial).*

- Reflexivity of parallel beta-reduction. Same unit, \ollabel{thm:refl} of the parallel-beta-reduction section (source line 671). Both N bredpar N and Q bredpar Q feed the contraction rule.
- Contraction rule of parallel beta-reduction. Same unit, \ollabel{defn:bredpar4} inside Definition defn:bredpar. Turns the pair of reflexive parallel steps into the parallel step over the contracted redex.

*Source reference.* Original definition defn-bredpar: sources/olp-lambda-calculus-church-rosser/pinned-original-source.tex, source lines 767–767. Prerequisite pointer invoked at proof line(s) 493; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original definition defn-bredpar4: sources/olp-lambda-calculus-church-rosser/pinned-original-source.tex, source lines 775–775. Prerequisite pointer invoked at proof line(s) 493; it names a standard fact of the same unit that the

counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original theorem thm-refl: sources/olp-lambda-calculus-church-rosser/pinned-original-source.tex, source lines 790–790. Prerequisite pointer invoked at proof line(s) 492; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

**Lemma 0.1.** *If $M =_\beta M'$, then $M \to_\beta M'$.*

*Proof.* If $M =_\beta M'$, then $M$ is $(\lambda[x][N])Q$, $M'$ is $N[Q/x]$, for some $x$, $N$, and $Q$. Since $N \to_\beta N$ and $Q \to_\beta Q$ by thm-refl, we immediately have $(\lambda[x][N])Q \to_\beta N[Q/x]$ by defn-bredpardefn-bredpar4. □